

Ivanna Aleman-Coronado

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EDUCATION

The University of Texas at Austin – Austin, TX | Robotics Honors Program

May 2029

B.S. Computer Science, Minor in Robotics

- Relevant Coursework: Computer Architecture, Autonomous Driving, Data Structures, Discrete Math, Linear Algebra, Probability & Statistics

TECHNICAL SKILLS

Languages: Python, C++, Java, C, JavaScript/TypeScript, HTML/CSS, Assembly

Robotics & Systems: ROS 2, ROS Navigation Stack, Boston Dynamics Spot SDK, Foxglove, Unity, Linux, Sensor fusion

ML/AI: Reinforcement Learning (PPO), Imitation Learning, Multi-agent systems, LLM-based evaluation, K-means/GMM clustering

Tools & Platforms: Git, VS Code, PyCharm, React, TACC, Figma, JFrame/JPanel

EXPERIENCE

Undergraduate Researcher, Autonomous Mobile Robotics Laboratory (AMRL)

Jan 2026 - Present

Austin, Texas

- **Robot Choreography Pipeline:** designed and implemented a ROS 2 system on a Boston Dynamics Spot robot to autonomously perform a fully synchronized 50-second dance routine, including a real-time audio beat-detection module and reusable custom move library.
- **Driver-Level debugging:** Diagnosed and resolved low-level failures in the spot_ros2 body-pose control stack, restoring reliable timed motion-command execution for the choreography system.
- **Autonomous vehicle RL research:** Built a Unity-Based coadaptive driving simulation with PPO training pipelines via PufferLib, distributed across TACC HPC Clusters, and a Qwen2.5-14B LLM behavior-scoring module feeding a multi-agent decision framework.
- **Critical bug fix:** traced and fixed a ROS 2 message-type mismatch in the lab's person-following tack with a targeted C++ patch to hdl_people_tracking_nodelet.cpp, restoring real-time tracking accuracy.

Machine Learning Research Intern, University of Texas at Dallas (StrALiNG Lab)

May 2024 - August 2024

Dallas, Texas

- **Anomaly Detection:** built K-means, Gaussian Mixture Model, and One-Class classifier pipelines to detect real-world denial-of-service attacks from network traffic.
- **Comparative evaluation:** benchmarked anomaly-based vs. signature-based detection across accuracy, recall, and precision, outperforming traditional signature-based baselines on novel attack patterns.
- Presented findings on ML-based network security to UTD faculty.

PROJECTS

Co-Adaptive Value Alignment for Autonomous Vehicles

Python, Unity, PPO, PufferLib, TACC HPC, Qwen2.5-14B

- Designed a multi-agent reinforcement learning environment in Unity simulating coadaptive merging behavior between human and autonomous drivers, training PPO policies at scale on TACC HPC clusters via PufferLib.
- Integrated a Qwen2.5-14B LLM as a behavior-scoring critic to evaluate driving safety/courtesy in dynamic race-track scenarios, feeding reward signals back into the multi-agent decision-making framework.

Spot "Dancing Queen" Autonomous Dance Demo

ROS 2, Python, Boston Dynamics Spot SDK

- Engineered an end-to-end ROS 2 pipeline that lets a Spot quadruped perform a hardcoded, fully autonomous 50-second choreography synced to music via a custom beat-detection algorithm.
- Built a modular move library and choreography state machine (dance_node.py / dance_moves.py), and debugged spot_ros2 body-pose driver issues to achieve smooth, precisely-timed motion.

Bit Blast Puzzle Game

Unity, C#, Figma

- 2D falling-block puzzle game teaching Boolean logic (AND/OR/NOT/XOR) through tile-matching gameplay; designed and shipped all core systems — spawning, collision, board evaluation, win/loss — in a 24-hour build sprint, deployed to Netlify.

Squirrel++ Game

Java, Java Swing, GIMP

- Drag-and-drop visual programming game where players build logic-driven programs (movement, IF/ELSE, state conditions) executed by a custom runtime interpreter, with a Java Swing UI supporting real-time editable conditional blocks.

LEADERSHIP & AWARDS

- **Hispanic Scholarship Fund Scholar** — selected from 70,000 applicants for academic achievement and leadership.
- **Hispanics in Computer Science Social Officer** — Organized all chapter social events.
- **NCWIT Aspirations in computing award recipient.**